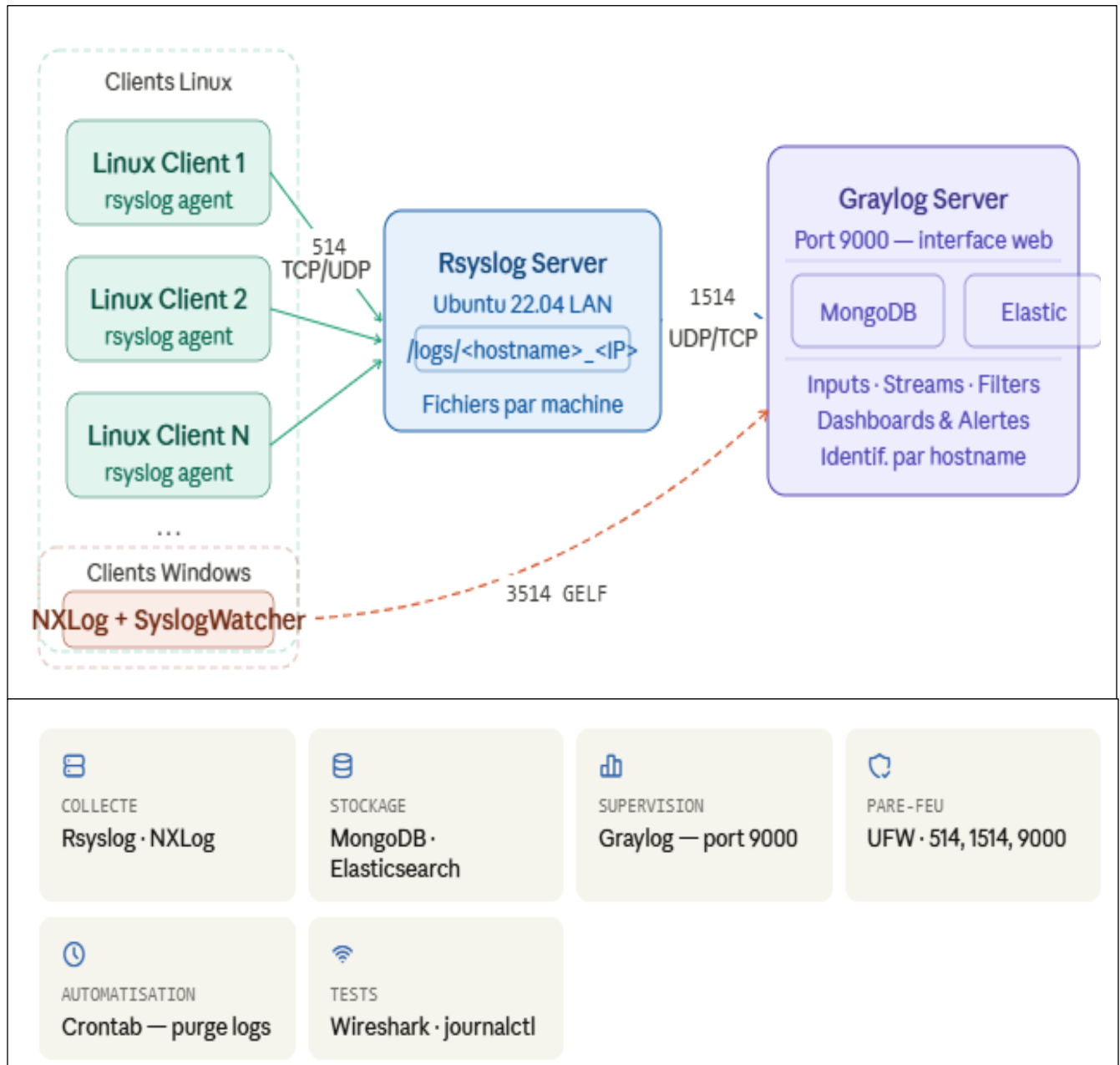


Description :

Schéma général de l'infrastructure comprenant le serveur Graylog, MongoDB, Elasticsearch, les clients Linux et Windows ainsi que les flux de logs centralisés.



Description :

Configuration du serveur Rsyslog et de Graylog pour la collecte et la centralisation des logs des clients Linux et Windows via les protocoles UDP / TCP / GELF.

```
root@serverubuntu-22
-----
OS: Ubuntu 22.04.5 LTS x86_64
Host: Virtual Machine Hypervisor
Kernel: 5.15.0-176-generic
Uptime: 4 hours, 30 mins
Packages: 740 (dpkg), 4 (snap)
Shell: bash 5.1.16
Resolution: 1024x768
Terminal: /dev/pts/0
CPU: Intel Ultra 7 265K (100%)
Memory: 7904MiB / 11055MiB
```

```
module(load="imuxsock") # provides support for local system logging
#module(load="immark") # provides --MARK-- message capability

# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

# provides TCP syslog reception
module(load="imtcp")
input(type="imtcp" port="514")
# log directory for Rsyslog & clients showing NTbios & IPs
template(name="PerHostLog" type="string"
          string="/logs/%HOSTNAME%_%FROMHOST-IP%.log")

*,* ?PerHostLog
# Rsyslog to Graylog web interface manager
*,* @@172.20.10.2:1514
```

```
<Extension _gelf>
  Module xm_gelf
</Extension>

<Input win>
  Module im_msvistalog
</Input>

<Output graylog>
  Module om_udp
  Host 172.20.10.2
  Port 3514
  OutputType GELF
</Output>

<Route graylog_route>
  Path win => graylog
</Route>
```

```
# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

*,* /logs/%HOSTNAME%.log
# provides TCP syslog reception
#module(load="imtcp")
#input(type="imtcp" port="514")
```

Description :

Contrôle de l'écoute du serveur Graylog/Rsyslog sur le port configuré pour la réception des logs.

```
root@serverubuntu-22:~# ss -tulnp | grep 1514
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=142))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=143))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=144))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=145))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=146))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=147))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=148))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=149))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=150))
udp UNCONN 0      0          *:1514    *:~ users:(("java",pid=52
29,fd=151))
```

Description :

Validation de l'ouverture des ports nécessaires à la collecte et à la centralisation des logs.

```
root@serverubuntu-22:~# ufw status verbose numbered
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
514 ALLOW IN Anywhere
514 ALLOW IN Anywhere
514/udp ALLOW IN 172.20.10.0/28
514/tcp ALLOW IN Anywhere
9000 ALLOW IN Anywhere
1514 ALLOW IN Anywhere
3514/tcp ALLOW IN Anywhere
3514/udp ALLOW IN Anywhere
514 (v6) ALLOW IN Anywhere (v6)
514 (v6) ALLOW IN Anywhere (v6)
514/tcp (v6) ALLOW IN Anywhere (v6)
9000 (v6) ALLOW IN Anywhere (v6)
1514 (v6) ALLOW IN Anywhere (v6)
3514/tcp (v6) ALLOW IN Anywhere (v6)
3514/udp (v6) ALLOW IN Anywhere (v6)
```

Description :

Capture et analyse du trafic réseau afin de vérifier la réception des logs sur le serveur.

```
root@serverubuntu-22:~# sudo tcpdump -i any port 3514
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 1500 bytes
15:05:06.557827 eth0 In IP 172.20.10.6.54247 > serverubuntu-22.3514: UDP
th 731
15:05:06.557864 eth0 In IP 172.20.10.6.54247 > serverubuntu-22.3514: UDP
th 744
15:08:13.326735 eth0 In IP 172.20.10.6.54247 > serverubuntu-22.3514: UDP
```

Description :

Contrôle de la création et du stockage des fichiers journaux reçus par le serveur.

```
root@serverubuntu-22: /
root@serverubuntu-22:/# ls -lh /logs/
total 8.0K
-rw-r----- 1 syslog syslog 6.1K Apr 23 22:25 %HOSTNAME%.log
root@serverubuntu-22:/# tail -5 /logs/%HOSTNAME%.log
2026-04-23T22:25:18.041402+00:00 serverubuntu-22 rsyslogd:
pid="3368" x-info="https://www.rsyslog.com"] start
2026-04-23T22:25:18.041743+00:00 serverubuntu-22 sudo: pam_
2026-04-23T22:25:24.594412+00:00 serverubuntu-22 sudo:
/bin/systemctl status rsyslog
2026-04-23T22:25:24.594564+00:00 serverubuntu-22 sudo: pam_
d=0) by user(uid=0)
```

Description :

Suppression automatique des fichiers journaux datant de plus de 7 jours afin d'optimiser l'espace disque du serveur.

```
# m h dom mon dow  command
0 0 */7 * * find /logs -type f -name "*.log" -mtime +7 -delete
```

Description :

Validation de la transmission des paquets Syslog entre les clients Linux/Windows et le serveur de centralisation.

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

udp == 3514

No.	Time	Source	Destination	Prot	Leng	Info
83	42.494840500	172.20.10.13	172.20.10.2	UDP	610	59375 → 3514 Len=568
84	42.494960800	172.20.10.13	172.20.10.2	UDP	623	59375 → 3514 Len=581
85	42.495289100	172.20.10.13	172.20.10.2	UDP	680	59375 → 3514 Len=638
86	42.495421400	172.20.10.13	172.20.10.2	UDP	626	59375 → 3514 Len=584
87	42.495569300	172.20.10.13	172.20.10.2	UDP	688	59375 → 3514 Len=646
88	42.495735500	172.20.10.13	172.20.10.2	UDP	775	59375 → 3514 Len=733
89	42.495835700	172.20.10.13	172.20.10.2	UDP	685	59375 → 3514 Len=643
90	42.495941100	172.20.10.13	172.20.10.2	UDP	685	59375 → 3514 Len=643

Protocol: UDP (17)

Header Checksum: 0x0000 [validation disabled]

[Header checksum status: Unverified]

Source Address: 172.20.10.4

Destination Address: 172.20.10.2

[Stream index: 6]

> User Datagram Protocol, Src Port: 49689, Dst Port: 3514

▼ Data (87 bytes)

Description :

Vérification de la réception et de l'affichage des logs dans l'interface Graylog.

